

Pushpender Singh Rathore

Cybersecurity Analyst · Malware Researcher ·
GSoC 2026 @ Metasploit

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GitHub github.com/Pushpenderrathore

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Second-year B.Tech Computer Science student at BRCM CET focused on **offensive security research, malware and botnet analysis**, and **low-level / systems programming** in C, C++, Assembly, and Python. **GSoC 2026 contributor** at the Metasploit Framework (Rapid7) building inline tracing presenters for Kerberos tickets and X.509 certificates. Hands-on experience across Active Directory abuse, ADCS ESC1-ESC16 exploitation, and reverse-engineering tooling.

EDUCATION

B.Tech, Computer Science & Engineering

Current · 2nd year

BRCM College of Engineering & Technology, Bahal, Haryana

Coursework and self-directed study in systems programming, networks, operating systems, and applied cryptography.

RSCIT Computer Diploma

Completed

Rajasthan State Certificate in IT · Government of Rajasthan

EXPERIENCE & OPEN SOURCE

GSoC 2026 Contributor — Metasploit Framework (Rapid7)

Jun – Aug 2026 (175 hrs)

Mentors: @jheysel-r7 (primary), @zeroSteiner (co-mentor)

- Designing and implementing **KerberosTicketTracePresenter** and **CertificateTracePresenter**: two new inline tracing presenters that bring HttpTrace-style transparency to Kerberos tickets and X.509 certificates inside msfconsole.
- Both presenter classes prototyped against AS-REQ / TGS-REQ flows in a Windows Server 2022 + ADCS lab on TEST.LOCAL; 13 + 14 RSpec examples written.
- Modelled on the existing HttpTrace architecture and the `krb5_ccache_presenter.rb` precedent for upstream-merge consistency.

Cybersecurity Internship — Unified Mentor & ICAT

Ongoing

Applied training in network security, vulnerability analysis, and ethical-hacking fundamentals; blockchain / distributed systems coursework at ICAT.

Open-Source Contributor — BlackArch Linux

Active

Working with the BlackArch penetration-testing distribution: package work, tool packaging, and offensive-security tooling research in a controlled environment.

FEATURED PROJECTS

PyScanner — Network Security Scanner

Python / C++

v10 with ~47 features. Includes a full C++ port with **zero-copy kernel TX** and a **stateless HMAC ISN** architecture. Roughly 37% Nmap feature parity with genuine advantages in batch transmission, offline CVE lookup, and the stateless ISN design.

Contactsd — Encrypted Contacts Daemon

C / Python

Secure CLI-based contacts manager. **AES-256-GCM** for confidentiality / integrity / authentication; keys derived from a master password via **PBKDF2-HMAC-SHA256** (150k iterations, per-record salt). All crypto via OpenSSL.

Venice Firewall — AI-Assisted Network Filter

Python / C / Shell

Integrates offline AI models and online AI services to analyse traffic patterns, detect anomalies, and resolve filtering decisions automatically. Real-time adaptive filtering with admin mail messaging.

MAC Address Randomizer Daemon

Python / Shell

Privacy-focused systemd service that periodically randomises MAC addresses for all active interfaces via macchanger. Arch / Debian / Fedora support, dynamic interface detection, and a kill-switch that blocks the network if spoofing fails.

TECHNICAL SKILLS

Languages: Ruby, Python, C / C++, Bash, x86 / x86-64 Assembly

Reverse Engineering: Ghidra, gdb, Binary Ninja, objdump, xxd, YARA

Pentesting: Metasploit, Nmap, Wireshark, Burp Suite, Impacket, Rubeus

CTF Platforms: HackTheBox, TryHackMe, OverTheWire, WeChall

AD & Auth: Kerberos, ADCS ESC1-ESC16, Kerberoasting, Pass-the-Hash, Golden / Silver Ticket

Cryptography: OpenSSL, ASN.1, DER / PEM, X.509, AES-GCM, PBKDF2

ML / Data: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, Transformers, Ollama

Tooling: Git, RSpec, pytest, systemd, Kali Linux, BlackArch

CERTIFICATIONS

- Cybersecurity Training — Unified Mentor
- Machine Learning — Cisco Networking Academy
- Blockchain — ICAT
- RSCIT Computer Diploma — Government of Rajasthan

PROFILES & LANGUAGES

GitHub Pushpenderrathore

Metasploit

fork Pushpenderrathore/metasploit-framework

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Portfolio pushpenderrathore.github.io

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Languages English, Hindi